



What one student costs us to run

Unit economics & running-cost analysis · July 2026 · all figures AUD, ex-GST

CONFIDENTIAL — INTERNAL**A\$0.23**

variable infrastructure cost per fully-active student, per year

A\$0.06of that is **all of Firebase** — reads, writes, functions, audio**A\$1.10**

all-in core cost per student at an 800-student platform

90%

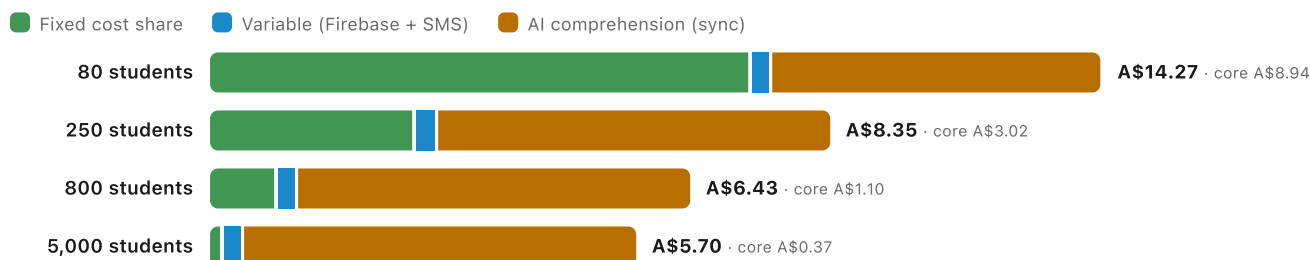
infrastructure gross margin at 800 students (A\$11 retail band)

The story in three points

- **The platform is extraordinarily cheap to run.** A student logging reading 200 nights a year — with a 60-second comprehension recording every single night — costs about **23 cents** of infrastructure. More than half of that is SMS sign-in messages, not servers.
- **Cost per student is a scale story, not a usage story.** Fixed platform costs (~A\$696/yr) dominate below ~500 students; they dilute from A\$8.70 to A\$0.14 per student as we grow from 80 to 5,000.
- **AI comprehension is the one real per-student cost decision.** The planned sync transcription pipeline costs A\$4.61 per student/year — ~20× everything else combined. A batch pipeline delivers the same feature for A\$0.86. This choice sets the margin on the A\$12 AI add-on: **56% vs 87%**.

Cost per student per year, by platform size

Upper-end usage · AI shown at the currently-planned sync transcription rate

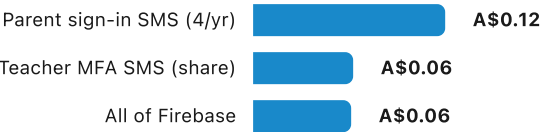


Core = fixed share + variable. Retail pricing is A\$15 / A\$13 / A\$11 per student/yr by school-size band (wholesale ~A\$7); core infrastructure margin runs 40% at a stand-alone 80-student platform, 77% at 250, 90% at 800, 95% at 5,000. The AI add-on is priced separately at A\$12 per student/yr.

Fixed costs dominate; Firebase is a rounding error

Variable cost per student-year — A\$0.23

200 logged nights + 200 recordings + full year of app use



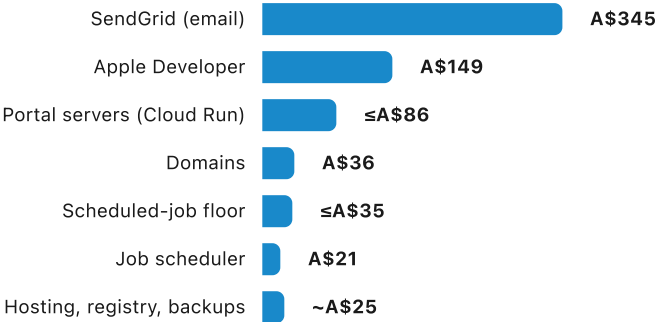
"All of Firebase" covers 24,300 database reads, 1,230 writes, 1,800 function runs, 96 MB of stored audio and its delivery — for six cents. SMS (A\$0.029 each) is now the single biggest variable line.

Free tiers make the real bill even smaller

Google's free daily quotas absorb most database usage below ~250–750 students, and the first 10 SMS/day are free (~600 students' worth). Until then the only line we genuinely pay is audio storage — Sydney buckets get no free storage tier.

Fixed platform costs — A\$696/yr

Independent of student count; dilute with scale



Everything is serverless and scales to zero — there is no idle server anywhere in the stack.

⚠ Half the fixed base is one email subscription

SendGrid's A\$345/yr Essentials plan (their free tier was retired in 2025) sends only a few hundred onboarding emails a month. Swapping to a cheaper transactional provider saves ~A\$250–300/yr — worth A\$3–4 per student at pilot scale.

Per-student summary (AUD per year, upper-end usage)

PLATFORM SIZE	FIXED SHARE	VARIABLE	CORE TOTAL	+ AI (SYNC)	+ AI (BATCH)	CORE MARGIN*
80 students (pilot school)	8.70	0.23	8.94	14.27	10.52	40%
250 students	2.78	0.23	3.02	8.35	4.60	77%
800 students	0.87	0.23	1.10	6.43	2.69	90%
5,000 students	0.14	0.23	0.37	5.70	1.96	95%

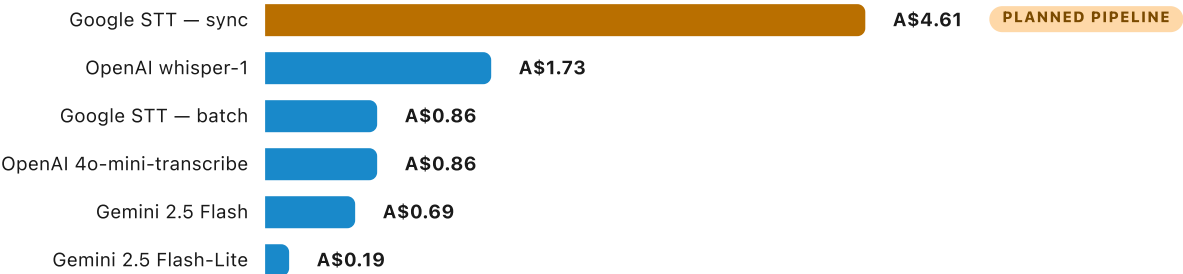
*Infrastructure margin against the applicable price band (A\$15 / 13 / 11 retail; ~A\$7 wholesale at 5,000). Excludes payment-channel costs — Apple's 15–30% commission on consumer subscriptions and the agreed partner percentage on the school channel — and GST treatment.

AI comprehension: pipeline choice sets the margin

Every recording transcribed: 200 × 60-second clips = 200 audio-minutes per student per year. Transcription cost is shown separately from everything else, per provider.

Transcription cost per student per year

Same feature, six ways to buy it — prices from official vendor pages, 15 Jul 2026



Plus the optional AI evaluation stage (Claude Haiku 4.5 reads each transcript and assesses comprehension): A\$0.72 per student/yr, or A\$0.36 via batch. Our modelled sync bundle (A\$5.33) independently matches the engineering estimate built into the AI pricing pitch (A\$5.20) within 2.5%.

Margin on the A\$12 AI add-on

PIPELINE	COST /YR	MARGIN
Sync STT + Haiku eval (as planned)	A\$5.33	56%
Batch STT + Haiku eval	A\$1.58	87%
Gemini Flash-Lite single-stage [†]	~A\$0.2–1.0	~92–98%

[†]The engineering backlog already lists both cheaper options: "STT batch recognition (~81% on the dominant cost line)" and a "Gemini Flash single-call bake-off". The overnight-report SLA absorbs batch latency — batch costs nothing in user experience.

Budget guard-rail

⚠ Our A\$100/month billing alert cannot absorb AI at sync rates

Today's entire Google bill is ~A\$16–30/month. Sync transcription adds **+A\$36/mo at 80 students, +A\$111/mo at 250, +A\$355/mo at 800**. Before enabling AI beyond a pilot school: raise the budget alert to match the expected AI spend — or ship the batch pipeline first (~81%).

The AI feature ships dark behind a kill switch (`platformConfig/aiEvaluation`) with per-school daily caps, so spend is controllable school-by-school from day one.

Why this is the decision that matters

- Transcription at sync rates is ~20× our entire remaining variable cost. No other engineering choice moves per-student economics by whole dollars.
- Batch vs sync is worth **A\$3.75 per student per year** — at 5,000 AI-enabled students, ~A\$18,750/yr of margin for the same feature.
- Provider choice is reversible: audio is stored in our bucket and pipelines are swappable; we are not locked in.

Five actions, in priority order

#	ACTION	WHY	IMPACT
1	Ship batch transcription before AI scales	Same feature, 81% cheaper; overnight SLA already absorbs the latency	A\$3.75 /student/yr
2	Raise the A\$100/mo billing alert before expanding the AI pilot	Sync AI spend alone exceeds the alert from ~250 students; today it fires only on real anomalies	avoids false "all-clear"
3	Replace SendGrid with a cheaper transactional-email provider	A\$345/yr plan for a few hundred emails/month — half our fixed base	~A\$250–300 /yr
4	Turn on audio retention (90–180 days)	Recordings currently accumulate forever — ~96 MB per student per year, compounding annually	caps storage growth
5	Enable the SMS rate limiter	SMS is now our largest variable cost and the only one an attacker can drive; the limiter exists but is switched off	closes abuse exposure

What we assumed (deliberately worst-case)

- 200 logged nights per student per year — every school night
- A maximum-length 60-second recording attached to every night
- Audio never deleted; every price at the top public tier, no free quotas counted
- 4 SMS sign-ins per parent/yr; a teacher MFA SMS every week
- All figures AUD ex-GST at A\$1.4404/US\$ (ECB, 14 Jul 2026)

Real bills will land **below** these numbers: typical students log fewer nights, recordings average under 60 s, and Google's free tiers absorb most database usage below a few hundred students.

How the numbers were built & checked

- Usage counts traced from the production code paths (every database operation a logged night actually triggers), not estimated
- Every vendor price fetched from official pricing pages on 15 Jul 2026 and independently re-verified
- Arithmetic, code claims, pricing and completeness each re-checked by separate adversarial review passes
- AI cost cross-checks the internal pricing-pitch model within 2.5%
- Modelled Google bill (A\$16–30/mo) is consistent with the live A\$100/mo budget alert never firing

Bottom line

Serving a family costs us cents; the business scales on fixed-cost dilution. The only material per-student spend decision is the AI transcription pipeline — choose batch, and the AI add-on carries an 87% margin while every other cost line stays under a dollar per student at scale.